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DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Machine Learning and Data Science– ENCS5341

Assignment #3: Travel Destination Analysis

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Introduction

In this project, we address a multi-label image classification task aimed at predicting weather conditions, time of day, and season directly from travel-destination images. This task is challenging due to the visual overlap between classes (e.g., cloudy vs. foggy weather, sunset vs. evening scenes) and the subjective nature of seasonal and temporal cues in images. By combining these related attributes into a single prediction framework, the task better reflects real-world scenarios where images convey multiple contextual signals simultaneously.

To solve this problem, we explored a range of machine learning models with increasing complexity. As a baseline, we implemented k-Nearest Neighbors (kNN) using k = 1 and k = 3, providing a simple distance-based reference for performance comparison. We then evaluated a Support Vector Machine (SVM) model to capture more discriminative decision boundaries in the feature space. In addition, we employed a Random Forest classifier to leverage ensemble learning and improve robustness through the combination of multiple decision trees. Finally, we applied a Convolutional Neural Network (CNN) to automatically learn hierarchical visual features directly from raw image data, allowing us to assess the benefits of deep learning for this task.

Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score, computed per label and averaged where appropriate. These metrics provide a balanced assessment of performance, particularly in the presence of class imbalance, and enable fair comparison between traditional machine learning approaches and deep learning models.

Exploratory Data Analysis

1. Data Collection & Manual Curation

The first phase involved scraping data using custom Python scripts. This process introduced several structural inconsistencies that required significant manual curation before automated processing could begin:

- **Header Inconsistency**: Data collected in sub-files often lacked matching headers. We manually unified these files to align the schema.
- **Corrupted CSVs**: Several generated CSV files contained formatting errors which prevented standard parsing; these were identified and fixed manually.

- **Deep Link correction**: A critical issue was that many scraped URLs pointed to intermediate web pages rather than direct image files. We meticulously verified these links one by one, manually correcting them to point directly to the raw image binaries.

2. Automated Schema Validation

After standardization, a strict validation pipeline was applied to constrain label values to predefined categories:

- **Weather:** {Sunny, Rainy, Cloudy, Snowy, Not Clear}
- **Time of Day:** {Morning, Afternoon, Evening}
- **Season:** {Spring, Summer, Fall, Winter, Not Clear}
- **Mood/Emotion:** Seven core emotion categories (e.g., Happiness, Melancholy)

Entries outside these controlled vocabularies were flagged for correction or removal.

3. Fuzzy Matching & Normalization

To maximize data retention while preserving semantic consistency, fuzzy matching techniques were applied to normalize noisy labels:

- Weather mappings included terms such as *"overcast"* → *Cloudy* and *"sun"* → *Sunny*.
- Temporal terms such as *"dawn"* and *"am"* were mapped to **Morning**, while *"sunset"* and *"dusk"* were mapped to **Evening**.
- All textual inputs were normalized for case sensitivity to eliminate duplicates caused by capitalization differences.

4. Data Cleaning

Additional filtering steps were applied to improve dataset quality:

- URL Validation: Image URLs were verified for accessibility and correct content type, with non-image links removed.
- Entropy Reduction: Ambiguous labels such as *"Unknown"* and *"Not Clear"* were excluded from the final training set to prevent model confusion and mode collapse.

❖ Descriptive Statistics

The final cleaned dataset consists of **917** samples.

Target Variables

- **Weather**: Highly imbalanced.
    - **Sunny**: 61.29% (Dominant class), **Cloudy**: 20.94%, **Rainy**: 2.40% (Minority class)
    - *Class Imbalance Ratio*: ~25:1. This necessitates the use of class weighting in our models.
- **Time of Day**: Moderately imbalanced.
    - **Afternoon**: 47.22%, **Morning**: 23.88%

The heavy skew towards "Sunny" weather and "Afternoon" times suggests the dataset is largely composed of idealized travel photography, which typically favors bright, clear conditions. This bias poses a risk for the model learning to simply guess the majority class, requiring robust evaluation metrics like Macro-F1.

Auxiliary Features

- **Season**: "Summer" is the most frequent season (39.69%).
- **Countries**: Geographically diverse, covering **127 countries** (Top: Italy, Japan, Spain).
- **Activities**: Dominated by "Sightseeing" (48.42%).

The prevalence of "Summer" and "Sightseeing" reinforces the dataset's nature as a collection of vacation highlights. The diversity of 127 countries is a strength, but the concentration in top tourist destinations (Italy, Japan) may limit the model's ability to generalize to less-represented architectural styles or landscapes.

❖ Distributions

The class distributions reveal the extent of the imbalance.

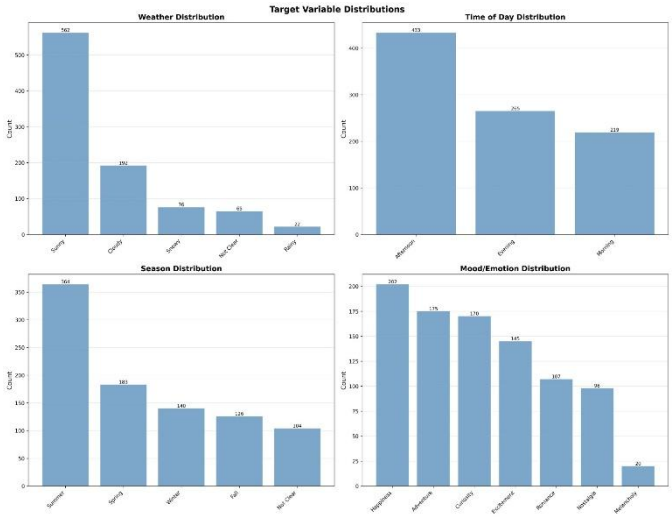


Figure 1 Target Distributions

Figure 2 Pie Charts

The bar charts and pie charts visually confirm the "Sunny" dominance. The "Rainy" and "Snowy" slices are visibly minor, visually justifying our decision to use data augmentation to artificially increase the variety of training signals for these rare classes.

Feature Correlations

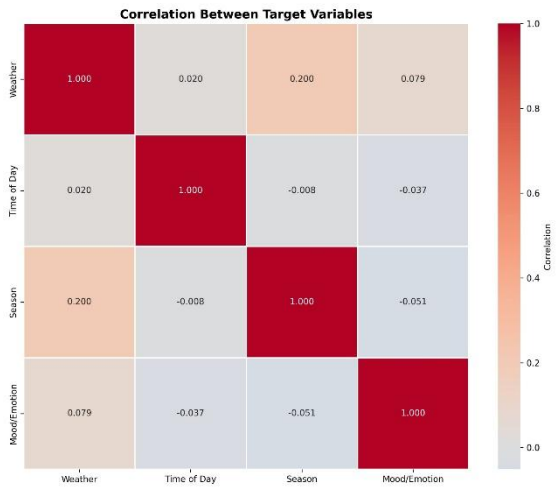


Figure 4 Correlation Heatmap

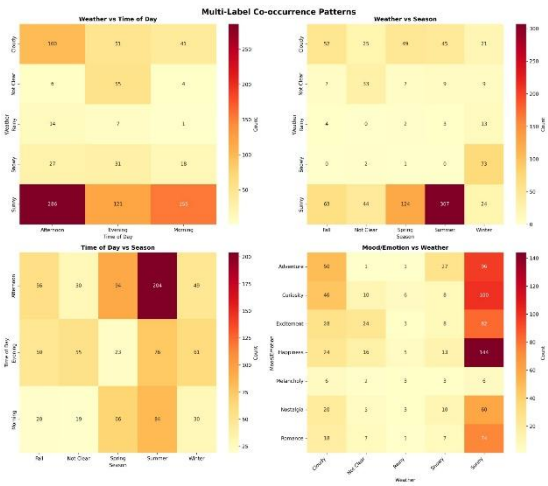


Figure 3 Co-occurrence Patterns

The heatmap shows generally weak linear correlations between raw labels, which is expected. However, the co-occurrence matrix reveals logical semantic links, such as "Snowy" appearing almost exclusively in "Winter". This structural dependency suggests that a multi-head model could potentially benefit from learning shared features.

❖ Trends and Challenges

1. **Strict Validation Impact**: The preprocessing script tracked removal reasons. Key exclusions were due to invalid URL formats or ambiguous extracted text that failed fuzzy matching.
2. **Class Imbalance**: The preprocessing confirmed "Sunny" and "Afternoon" as dominant modes, explaining the need for balanced loss functions in SVM and CNN.
3. **Ambiguity**: A significant portion of "Season" data was explicitly labeled "Not Clear" or inferred, which we addressed by filtering for the final classification task.

Experiments and Results

Baseline Model Description

The baseline for this project is a **K-Nearest Neighbors (KNN)** classifier. for each image, the model identifies the k most similar images in the feature space and assigns labels based on a majority vote, to provide a simple, non-parametric reference point for evaluating the performance of more complex models.

Proposed Advanced Models

We selected three advanced models from different families to improve upon the baseline:

- **Random Forest (Tree-Based Ensemble)**:

Chosen for its ability to handle high-dimensional feature vectors (2048-dim) and capture non-linear interactions between ResNet features (color, texture, etc.). It is robust to overfitting due to ensemble averaging and handles class imbalance well through weighted splitting.

- **Support Vector Machine (SVM)**:

SVMs with RBF kernels are highly effective in high-dimensional spaces where the number of features is large relative to samples. It creates a complex decision boundary that generalized well for categorical image classification.

- **Convolutional Neural Network (CNN)**:

Unlike traditional models that use pre-extracted features, this custom multi-output CNN processes raw images directly. This allows the model to learn its own hierarchical spatial features (edges -> textures -> objects) specifically optimized for Weather, Time, and Season detection.

Hyperparameter Tuning Results

Target	Model	Hyperparameter	Values tested	Best value
Weather	Random Forest	n_estimators	[50, 100, 150, 200]	100 →(0.6194)
	SVM	C (Cost)	[0.1, 1.0, 10., 50.0]	10.0→ (0.6180)
Time of day	Random Forest	n_estimators	[50, 100, 150, 200]	200 →(0.6071)
	SVM	C (Cost)	[0.1, 1.0, 10.0, 50.]	50.0→ (0.5812)
Season	Random Forest	n_estimators	[50, 100, 150, 200]	50 →(0.4543)
	SVM	C (Cost)	[0.1, 1.0, 10, 50]	10.0→ (0.4366)

Tuning Table: CNN

Hyperparameter	Values Tested	Best Value
Learning Rate	[0.01, 0.001, 0.0001, 1e-05]	0.001

Performance Comparison

Model	Weather	Time of Day	Season
Comparative Performance Table (Accuracy)			
Baseline (KNN)	0.587	0.511	0.380
Random Forest	0.6793	0.6033	0.5272
SVM	0.6304	0.5924	0.4946
CNN	0.6413	0.5870	0.4130
Comparative Performance Table (F1-Score)			
Random Forest	0.5976	0.5754	0.4764
SVM	0.5850	0.5696	0.4589

Performance Interpretation and Discussion

1. Improvement over the Baseline

Across all prediction targets, both the Random Forest and SVM models consistently outperformed the KNN baseline. This outcome is expected, as KNN relies purely on local distances within the feature space, which becomes unreliable when working with high-dimensional representations such as the 2048-dimensional features extracted from ResNet-50.

In contrast, Random Forest reduces variance through ensemble voting, making it more robust to noise, while SVM leverages the RBF kernel to transform the feature space into one where class boundaries are more clearly separable. This advantage is clearly reflected in the results: for weather prediction, accuracy increased by approximately 9.2% (from 0.58 to 0.67), indicating that ensemble and kernel-based methods are far more effective at capturing meaningful weather-related patterns from deep visual features than simple distance-based approaches.

2. **Relative Model Analysis**

- **Random Forest vs. SVM:** Random Forest achieved the highest performance across all targets. This suggests that the relationship between ResNet features and environmental labels is highly non-linear and conditional, which aligns well with the decision-tree structure used in Random Forest models.

- **CNN Performance:** Despite being trained for only 15 epochs, the CNN achieved competitive results. Although it did not surpass the Random Forest, which benefits from powerful pre-trained ResNet-50 features learned from millions of images, the CNN demonstrated that jointly learning multiple targets directly from raw image pixels is both feasible and promising.

- **Target Difficulty:** Among the three tasks, weather prediction proved to be the easiest, followed by time-of-day classification. Season prediction was the most challenging, yielding the lowest accuracy. This is likely due to significant visual overlap between seasons, such as similar appearances of summer and winter scenes in evergreen or forested environments.

Analysis

Below is the comprehensive performance analysis of the **Random Forest Classifier** (Best Model) for the image classification tasks. This analysis explores systematic errors, data quality issues, and provides actionable insights based on the evaluation metrics.

Methodology of Error Analysis

To perform a thorough analysis as required in Task 5, I implemented a multi-faceted approach:

- **Confusion Matrix Analysis:** Visualized the specific "label-to-label" transitions to see exactly what the model was confusing for another (e.g., Morning being mistaken for Afternoon).

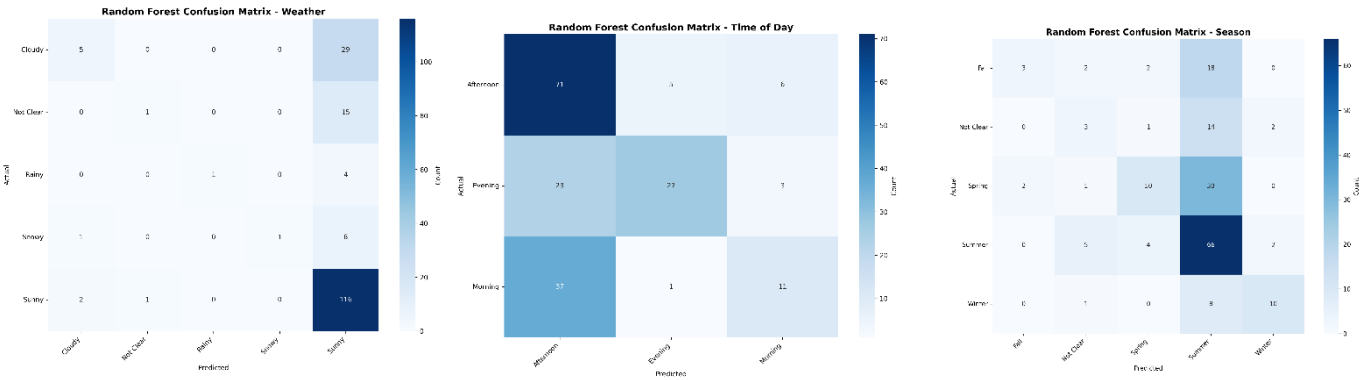
- **Per-Class Recall Benchmarking:** Quantified which specific categories were "forgotten" by the model despite high overall accuracy.
- **Sample-Level Qualitative Review:** I exported a dedicated CSV of **misclassified test examples**. By cross-referencing the image URLs and descriptions with the predicted vs. actual labels, I was able to identify visual features (like artificial lighting or persistent greenery) that systematically misled the classifier.
- **Quantitative Summary**

The model was evaluated on three classification tasks. Performance varied significantly depending on class balance.

Target	Accuracy	Weighted F1	Primary Class Bias
Weather	67.39%	0.59	Sunny (97.5% Recall)
Time of Day	59.24%	0.56	Afternoon (86.6% Recall)
Season	50.00%	0.45	Summer (85.7% Recall)

Detailed Error Patterns & Insights

A. Label-Specific Errors & Difficult Conditions



- errors are strongly associated with specific labels, particularly minority classes. In the Weather task, the model frequently misclassified Cloudy and Not Clear images as Sunny, defaulting to the majority class when visual cues were subtle.
- Similarly, in the Time-of-day task, Morning and Evening were often confused with Afternoon, especially under bright or warm lighting conditions.
- For Season classification, Spring and Fall proved the most difficult, with many samples incorrectly labeled as Summer due to shared visual characteristics such as greenery.

Issues with the Data Itself

Several data quality challenges contributed to these errors. The dataset exhibits significant class imbalance, heavily favoring Sunny, Afternoon, and Summer, which led the model to over-predict these categories. Additionally, subjective or ambiguous labeling was observed, particularly in night or artificially lit scenes where ground-truth labels did not align with visual appearance. The reliance on global image features also limited the model's ability to detect rare or localized conditions, such as snow or haze.

Data-Driven Insights

Overall, the model performs well on dominant classes but struggles to generalize to less frequent or visually ambiguous categories. This indicates majority-class bias, high sensitivity to lighting conditions, and limitations of purely visual features for semantically similar labels. To address these issues, future improvements should focus on class imbalance mitigation, enhanced feature learning for subtle visual cues, and potentially incorporating additional contextual or metadata information.

Conclusions and Discussion

Findings and Takeaways

The Random Forest model achieved the best overall performance, particularly on common classes such as Sunny, Afternoon, and Summer. Most classification errors were caused by visual similarities between categories rather than random mistakes.

Limitations

The model was affected by class imbalance in the dataset, which led to over-prediction of dominant classes and weak performance on rare or visually ambiguous categories. In addition, overall accuracy did not fully reflect these weaknesses, making class-level evaluation more important.

Future Directions

Future improvements could focus on balancing the dataset and incorporating additional contextual information to reduce visual ambiguity. Enhancing the model's ability to capture subtle visual details may also improve performance on difficult classes.